Manish Chhetri

0449 863 141 | chhetrim901@gmail.com | linkedin.com/in/manish-chhetri1 | mchhetri.vercel.app | github.com/manishchh

Aspiring Software Engineering graduate with experience in full-stack development, AI applications, and secure software design. Skilled in React, Node.js, Python, and JWT authentication, with hands-on experience delivering practical solutions through research and personal projects. Seeking a graduate software engineering role to apply strong problem-solving, technical, and leadership skills.

Work Experience

Honours Research Project | University of South Australia — Aug. 2024 – Jun. 2025

- Developed an AI powered grading application with the **Centre for Change and Complexity in Learning (C3L)** to support automated assessment evaluation and feedback generation.
- Integrated an LLM-powered grading workflow into an existing Django/Python web application using LangChain by **collaborating with developers and academic stakeholders** to gather requirements and refine the solution.
- Achieved **60% grading accuracy** improvement through iterative prompt engineering and validation against real assessment datasets, reducing manual grading workload for academic staff.

Projects

Chatify | JavaScript, React, Node.js — Jan. 2026 – Feb. 2026

- Built a full-stack real time messaging web application using Socket.IO, RESTful APIs, and MongoDB-backed message storage, enabling instant communication and reliable offline access to conversation history.
- Strengthened application security by implementing **JWT** authentication, **bcrypt** password hashing, request rate limiting, and **Arcjet bot detection** across sign-up, sign-in, and public API flows.
- Built a responsive front-end with Tailwind CSS and Zustand, while integrating Cloudinary media uploads and **validating endpoints** through Postman to support robust cross-device functionality.

COVID-19 Data Analysis & Forecasting Dashboard | R, Shiny, rvest, tidyverse — Apr. 2025

- Built a reusable data pipeline in R to scrape, clean, and prepare COVID-19 case and death data across four Australian states from publicly available web sources.
- Analysed state level trends using population-adjusted comparisons and **regression modelling** and communicated findings through an interactive Shiny dashboard and written report.

Patient Vital Management System | C++ — May. 2024

- Developed a patient vitals management system to record health measurements, evaluate disease-specific alert levels, and trigger critical hospital and GP notifications for high-risk patients.
- **Reduced code duplication by 40%** and improved system flexibility by applying Strategy, Composite, Observer, and Adapter design patterns, enabling rapid addition of new disease types without core system modifications.

Country Emission | C#, ASP.NET Core MVC — Oct. 2023

- Developed a database-backed web application to manage and display country emissions data, implementing controller logic, Razor views, and **Entity Framework** to support structured data entry, retrieval, and maintenance.
- Optimized data retrieval performance using **LINQ queries** and **SQL Server** indexing strategies, while implementing View Models and Bootstrap responsive design to deliver a maintainable, user-friendly emissions tracking interface.

Education

University of South Australia — Oct. 2025

Bachelor of Software Engineering (Honours)

Relevant Coursework: Data Structures and Algorithms, Object Oriented Programming, AI and Machine Learning, Agile Development, System Architecture, Software Development Life Cycle (SDLC), Secure Coding Practices and IT Governance, Design Patterns, Cisco Networking Fundamentals

Advanced Electives: Data Science and Visualisation *(Postgraduate Course)*

Leadership & Activities

Competitive Programming Club — Feb. 2025 – Present

- Competed in contests such as LeetCode, Codeforces, and inter-university competitive programming rounds.
- Collaborated with teammates to optimize solutions under time constraints, strengthening problem-solving methodology, algorithmic thinking, and technical communication skills.

Uni Mentor Feb. 2024 – Nov. 2024

- Guided **40+ first-year students** through university transition by providing academic support, campus resources navigation, and peer mentoring during orientation and semester activities.
- Fostered an inclusive environment through active engagement strategies, resulting in increased student participation and positive feedback from program coordinators.

Technical Skills

Programming: Python, Java, C++, JavaScript, HTML/CSS, SQL, R
Frameworks & Libraries: React, Node.js, Express.js, Django, Zustand, LangChain, Pandas, NumPy
Tools: Git, GitHub Actions, Postman, Jira, Figma
Development Practices: Agile, Scrum, Unit Testing, SOLID Principles, Design Patterns

Certifications

HackerRank: JavaScript, Node.js, SQL *(Intermediate)* Issued: Apr. 2025